



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI SINGARAVELAN TRADING COMPANY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

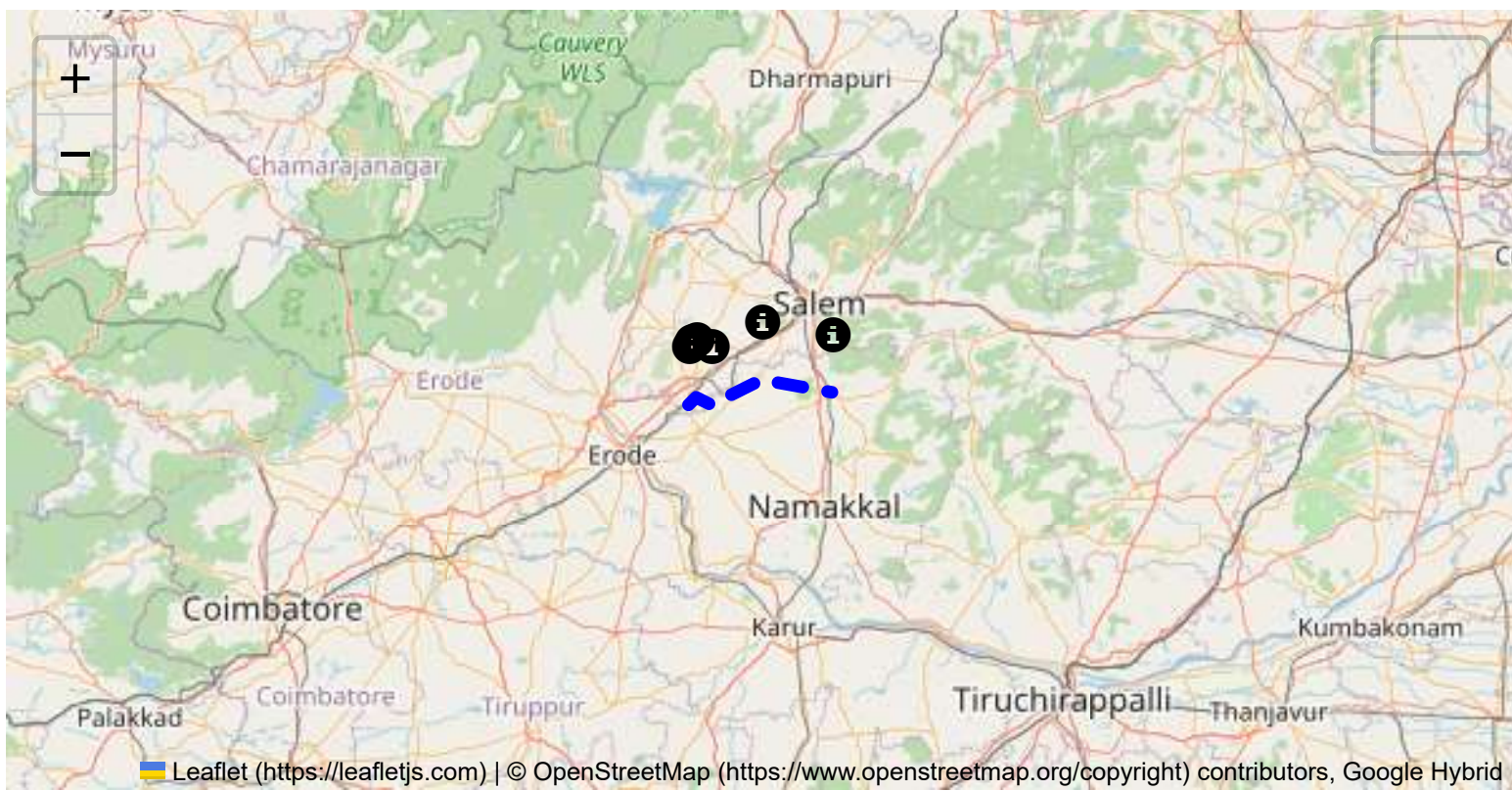
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

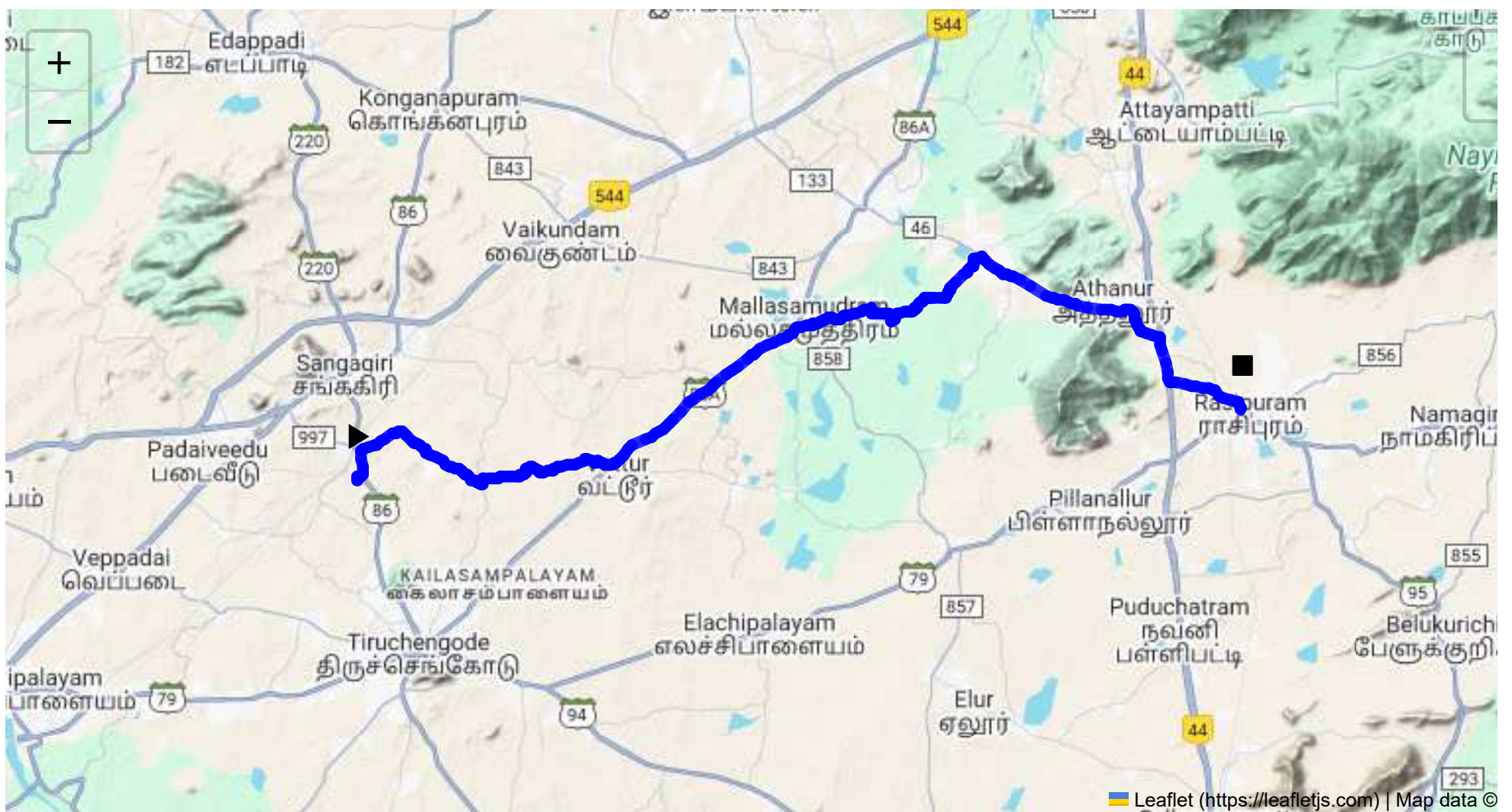
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.40 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.462549, 78.18232)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Rasipuram via the specified locations covers approximately 43.40 kilometers. The journey traverses through multiple rural and semi-urban areas, which typically display a mix of narrow roads, market zones, and residential stretches.

Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with seasonal monsoons. From June to September, the southwest monsoon can bring heavy rainfall, potentially leading to slippery roads and reduced visibility. October to December sees the northeast monsoon, which may also cause similar challenges. Extreme heat during the summer months can affect vehicle performance.

Traffic Patterns

The route passes through busy market and towns areas, notably at Sankari R.S and Mallasamudram. Peak hours are typically between 8:00-10:00 AM and 5:00-7:00 PM, with increased congestion due to local traffic and school commuting times. Expect potential slowdowns around these areas.

Road Quality and Infrastructure

Road quality varies along the route. Areas near towns may have better-maintained roads, but expect uneven surfaces and potential potholes in less frequently traveled rural stretches. Infrastructure such as signage and lighting may be lacking in remote areas, particularly after Kozhikkalnatham.

Alternative Routes for Emergencies

In case of emergencies, consider using the NH44 highway between Sankagiri and Rasipuram for a larger, more accessible route. This highway offers better road conditions and quicker access to emergency services.

Local Regulations on Hazardous Material Transport

In Tamil Nadu, there are strict regulations regarding the transportation of hazardous materials. Permits are required, and routes must minimize risks to populated areas. Be prepared for inspections and maintain all regulatory documentation onboard.

Historical Incidents

While specific incident data is limited, rural Tamil Nadu has seen road accidents involving heavy vehicles due predominantly to narrow roads and driver fatigue. However, notable incidents involving hazardous materials are relatively rare.

Environmental Considerations

The route passes through agricultural areas, necessitating careful driving to prevent any hazardous material spills that could affect soil and water sources. Wildlife and livestock crossings are possible in rural sectors, requiring cautious navigation.

Communication Coverage

Communication networks along the route are generally reliable, but signal strength can weaken in more remote sections between towns. It is advisable to equip vehicles with both mobile and VHF radios for

uninterrupted communication.

Emergency Response Times

- In town areas: Expect an average response time of 15-20 minutes due to proximities to local fire stations or police outposts.
- In rural areas: Response times may be extended to 30-45 minutes due to distance from services and challenging road access.

Overall Risk Assessment Summary

The route from Sangagiri to Rasipuram is a mix of urban and rural driving conditions, with moderate risk due to variable road quality, seasonal weather challenges, and occasional traffic congestion. Key risks include potential delays, road surface issues, and limited emergency access in rural sections. Proper preparation, adherence to regulations, and maintaining communication will mitigate risks effectively. Always have emergency contacts and alternative strategies ready in case of unforeseen incidents.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.44194, 77.90923	30 KM/Hr
5	Blind Spot	Blind Spot	11.43673, 77.91703	10 KM/Hr
6	Turn	High	11.43919, 77.91702	15 KM/Hr
7	Turn	Medium	11.43930, 77.91729	30 KM/Hr
8	Turn	Medium	11.43931, 77.92291	30 KM/Hr
9	Turn	Medium	11.43996, 77.92574	30 KM/Hr
10	Turn	Medium	11.43982, 77.93018	30 KM/Hr
11	Turn	Medium	11.44033, 77.93076	30 KM/Hr
12	Turn	Medium	11.44071, 77.93211	30 KM/Hr
13	Turn	Medium	11.44096, 77.93223	30 KM/Hr
14	Turn	Medium	11.44135, 77.93220	30 KM/Hr
15	Turn	Medium	11.44175, 77.93249	30 KM/Hr
16	Turn	High	11.44248, 77.93247	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.44289, 77.93420	30 KM/Hr
18	Turn	Medium	11.44547, 77.95309	30 KM/Hr
19	Turn	High	11.49060, 78.02684	15 KM/Hr
20	Blind Spot	Blind Spot	11.49133, 78.03257	10 KM/Hr
21	Turn	High	11.49181, 78.03258	15 KM/Hr
22	Turn	Medium	11.49362, 78.03714	30 KM/Hr
23	Turn	Medium	11.49411, 78.03873	30 KM/Hr
24	Turn	Medium	11.49379, 78.04238	30 KM/Hr
25	Turn	Medium	11.49631, 78.04976	30 KM/Hr
26	Turn	High	11.49672, 78.05001	15 KM/Hr
27	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
28	Turn	Medium	11.49634, 78.05316	30 KM/Hr
29	Turn	Medium	11.49557, 78.05334	30 KM/Hr
30	Turn	High	11.49546, 78.05344	15 KM/Hr
31	Turn	Medium	11.49524, 78.05507	30 KM/Hr
32	Turn	Medium	11.49544, 78.05546	30 KM/Hr
33	Turn	Medium	11.49528, 78.05693	30 KM/Hr
34	Turn	Medium	11.49547, 78.05727	30 KM/Hr
35	Blind Spot	Blind Spot	11.49549, 78.05900	10 KM/Hr
36	Turn	High	11.49470, 78.05907	15 KM/Hr
37	Turn	Medium	11.49464, 78.05915	30 KM/Hr
38	Turn	High	11.49454, 78.05993	15 KM/Hr
39	Turn	High	11.49306, 78.05985	15 KM/Hr
40	Turn	Medium	11.49304, 78.06006	30 KM/Hr
41	Turn	High	11.49368, 78.06138	15 KM/Hr
42	Turn	High	11.49557, 78.06189	15 KM/Hr
43	Turn	High	11.49662, 78.06786	15 KM/Hr
44	Turn	High	11.49734, 78.06795	15 KM/Hr
45	Turn	High	11.49740, 78.06877	15 KM/Hr
46	Turn	High	11.49905, 78.06933	15 KM/Hr
47	Turn	Medium	11.49960, 78.07110	30 KM/Hr
48	Turn	High	11.50121, 78.07189	15 KM/Hr
49	Turn	Medium	11.50068, 78.07327	30 KM/Hr
50	Turn	Medium	11.50094, 78.07390	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
51	Turn	Medium	11.50086, 78.07437	30 KM/Hr
52	Turn	High	11.50082, 78.07865	15 KM/Hr
53	Turn	High	11.50378, 78.08052	15 KM/Hr
54	Turn	High	11.50435, 78.08045	15 KM/Hr
55	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
56	Turn	High	11.51417, 78.09090	15 KM/Hr
57	Turn	Medium	11.51442, 78.09104	30 KM/Hr
58	Turn	Medium	11.51466, 78.09146	30 KM/Hr
59	Blind Spot	Blind Spot	11.51500, 78.09169	10 KM/Hr
60	Turn	High	11.48720, 78.15384	15 KM/Hr
61	Turn	High	11.47225, 78.15722	15 KM/Hr
62	Turn	Medium	11.46725, 78.17485	30 KM/Hr
63	Turn	High	11.46506, 78.18142	15 KM/Hr
64	Turn	Medium	11.46287, 78.18228	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium
6	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
7	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
8	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
9	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
10	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	school	SRV Hi tech matric hr secondary school	11.4730143, 78.1538356	30 km/h	Medium
5	school	SRV School	11.4688166, 78.1634582	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44861, 77.87429



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44194, 77.90923



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43673, 77.91703



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43919, 77.91702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43930, 77.91729



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43931, 77.92291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43996, 77.92574



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43982, 77.93018



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44033, 77.93076



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44071, 77.93211



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44096, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44135, 77.93220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44175, 77.93249



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44248, 77.93247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44289, 77.93420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44547, 77.95309



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49060, 78.02684



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49133, 78.03257



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49181, 78.03258



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49362, 78.03714



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49411, 78.03873



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49379, 78.04238



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49631, 78.04976



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49672, 78.05001



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49634, 78.05316



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49557, 78.05334



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49546, 78.05344



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49524, 78.05507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49544, 78.05546



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49528, 78.05693



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05727



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49549, 78.05900



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49470, 78.05907



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49464, 78.05915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49454, 78.05993



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49306, 78.05985



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49304, 78.06006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49368, 78.06138



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49557, 78.06189



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49662, 78.06786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49734, 78.06795



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49740, 78.06877



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49905, 78.06933



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49960, 78.07110



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50121, 78.07189



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50068, 78.07327



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50094, 78.07390



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50086, 78.07437



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

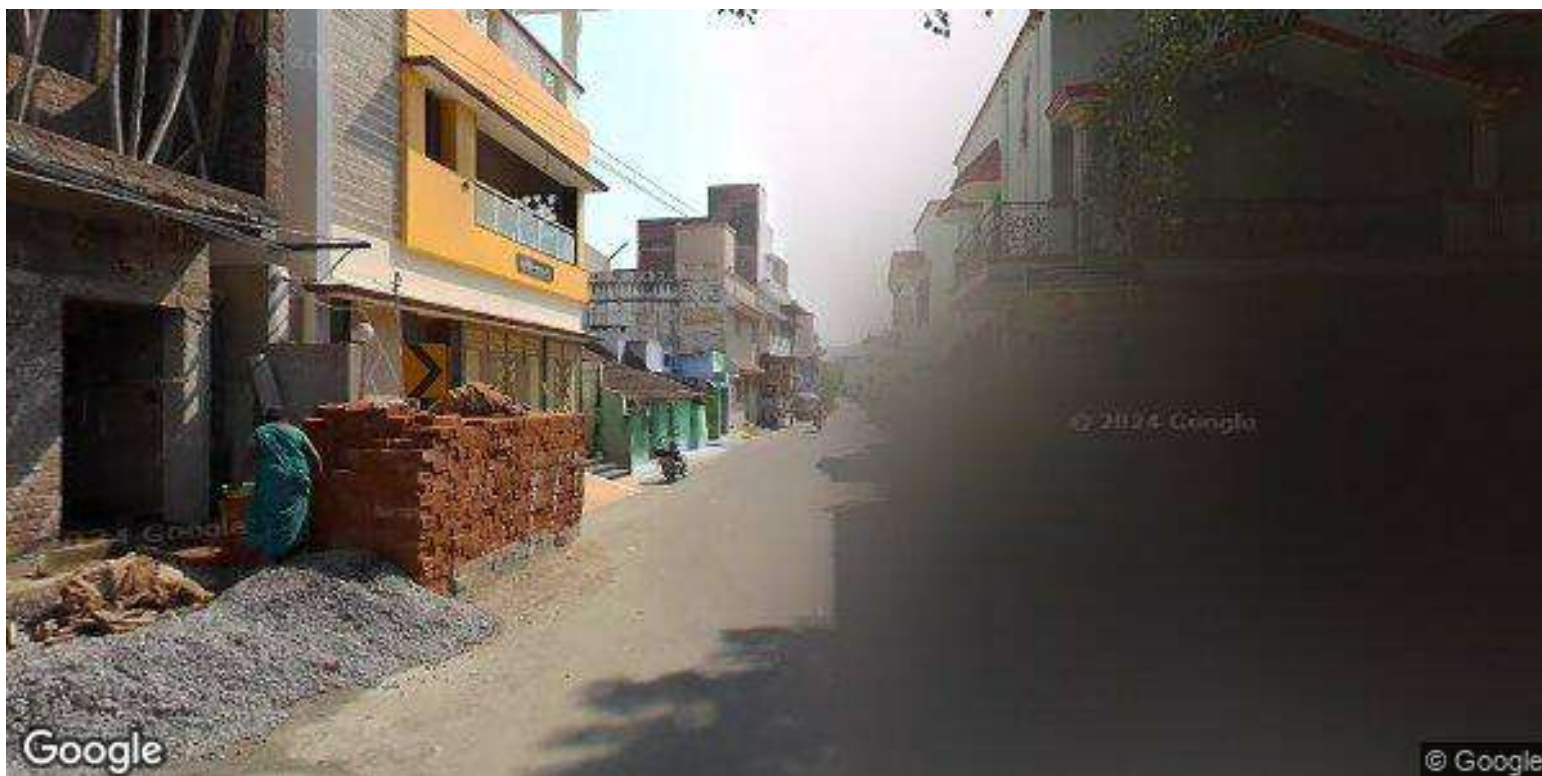
Coordinates: 11.50082, 78.07865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50378, 78.08052



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50435, 78.08045



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.51417, 78.09090



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51442, 78.09104



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51466, 78.09146



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.51500, 78.09169



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48720, 78.15384



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47225, 78.15722



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46725, 78.17485



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46506, 78.18142



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46287, 78.18228
